

INSTALL X-RAY TUBE AND HEAT EXCHANGER

This section explains how to install the X-ray tube and heat exchanger on systems where these items were shipped separately.

Required Tools:

This process requires the following tools, which may be in addition to your standard tool kit:

9/16" Socket, Ratchet or a closed-end wrench

5/32" Hex Key

Hoist Assembly (7" tube beam installation)

Required Materials:

Transformer oil

Protective cover for Patient Support couch top.

INSTALLATION NOTES, WARNINGS AND CAUTIONS



WARNING

USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANEL IS OPEN. THESE PANELS HAVE SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY.



WARNING



ELECTROCUTION HAZARD. USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GATNRY COVERS ARE REMOVED. CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

 WARNING

CRUSH HAZARD. SCISSOR-ACTION MACHINERY IS EXPOSED WHEN THE PATIENT SUPPORT COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO DO SO CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND SERIOUS INJURY OR DEATH TO PERSONNEL.

XRAY TUBE COMPONENT WEIGHTS		
Tube Type	Tube With Cradle	Heat Exchanger
7 INCH	132 pounds (60 kg)	45 pounds (20.5 kg)

 CAUTION

ROTATE THE SCAN FRAME COUNTER-CLOCKWISE ONLY. CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES.

INSTALL X-RAY TUBE AND HEAT EXCHANGER

At this time:

The front and rear upper Gantry covers are open and the rear Gantry covers are off.

The scan frame should be pinned at the 12 o'clock position. Refer to Figure 46.

System power is off at the wall box.

Prepare Gantry and Patient Support for Installation of Tube and Heat Exchanger

WARNING

MAKE NO ATTEMPT TO REMOVE THE HIGH VOLTAGE FEDERAL CONNECTORS FROM THE X-RAY TUBE. THEY ARE PERMANENTLY ATTACHED AND STAY WITH THE TUBE. REMOVING THESE CONNECTORS WILL CAUSE SERIOUS DAMAGE AND/OR DESTRUCTION OF THE TUBE.

1. Assemble the hoist and secure it to the Patient Support couch top. Screw the *knob bolts* in securely and make sure the Clamp Base seats firmly in position on the Patient Support. See Figure 47.

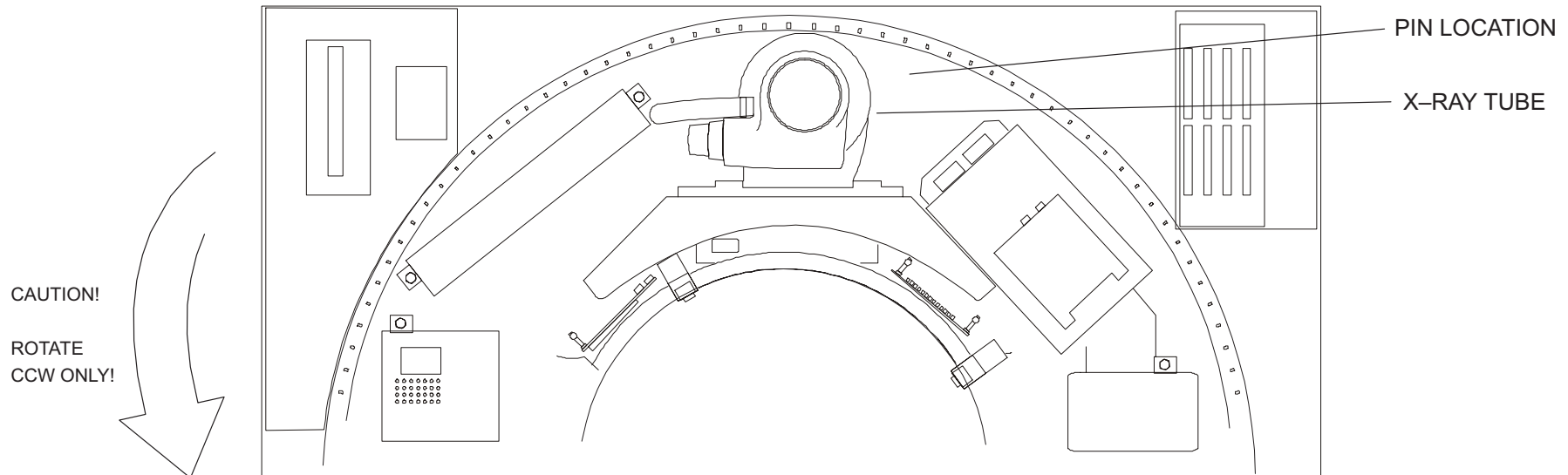


FIGURE 46 SCAN FRAME PINNED WITH TUBE AT 12 O'CLOCK

2. Place a protective cover on the Patient Support (such as a towel or several layers of a sheet).
3. Remove the four Cradle Locking Hex Head Bolts enough to slide Mounting Rails against Cradle and Tube Shelf. Install the Tray Table Slide to the Tube Shelf. Refer to Figure 48.
4. Remove the Cradle-locking Hex Head Bolts and Washers. Refer to Figure 49. (This figure shows the tube in its installed position.)

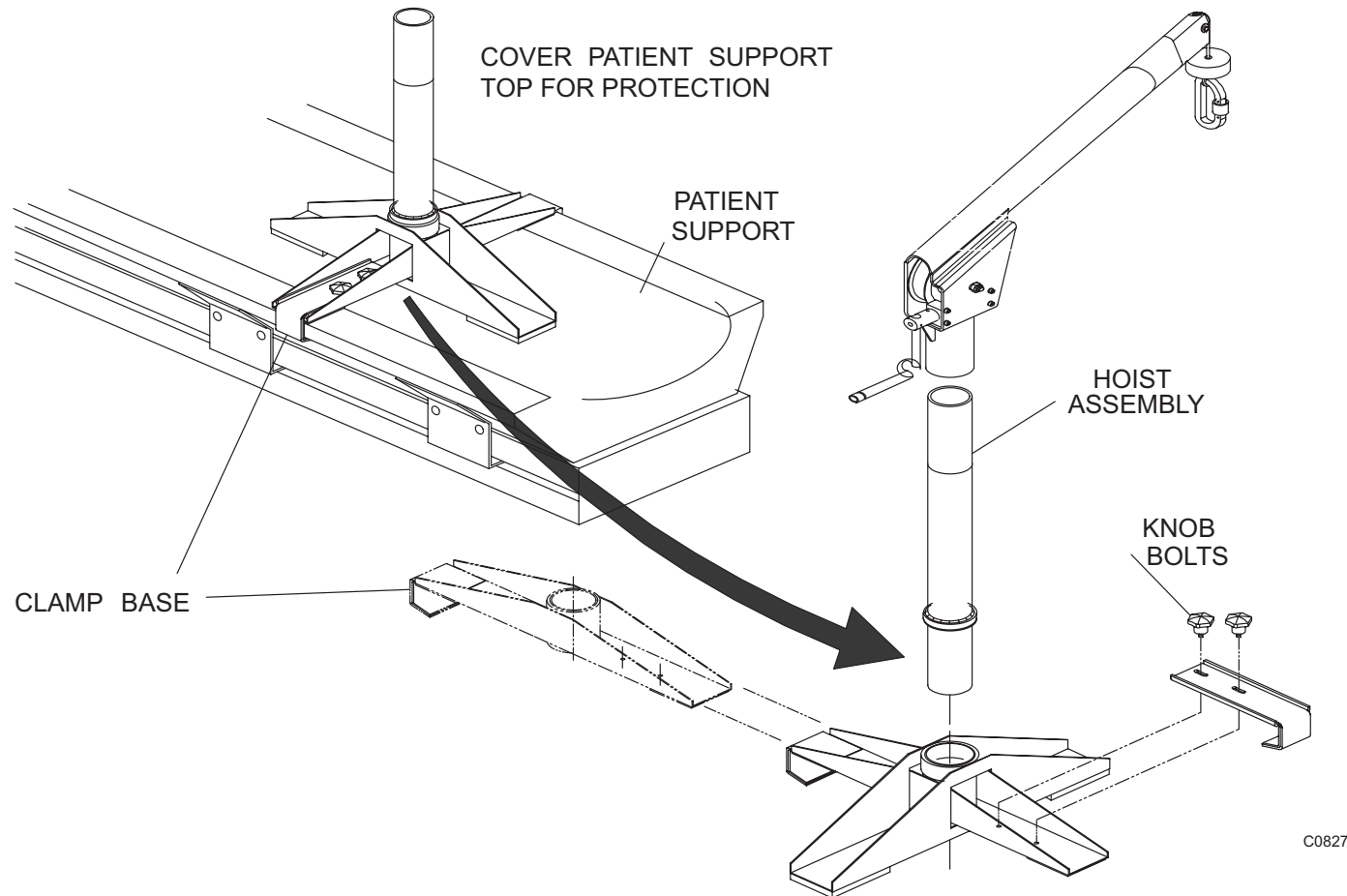


FIGURE 47 HOIST ASSEMBLY

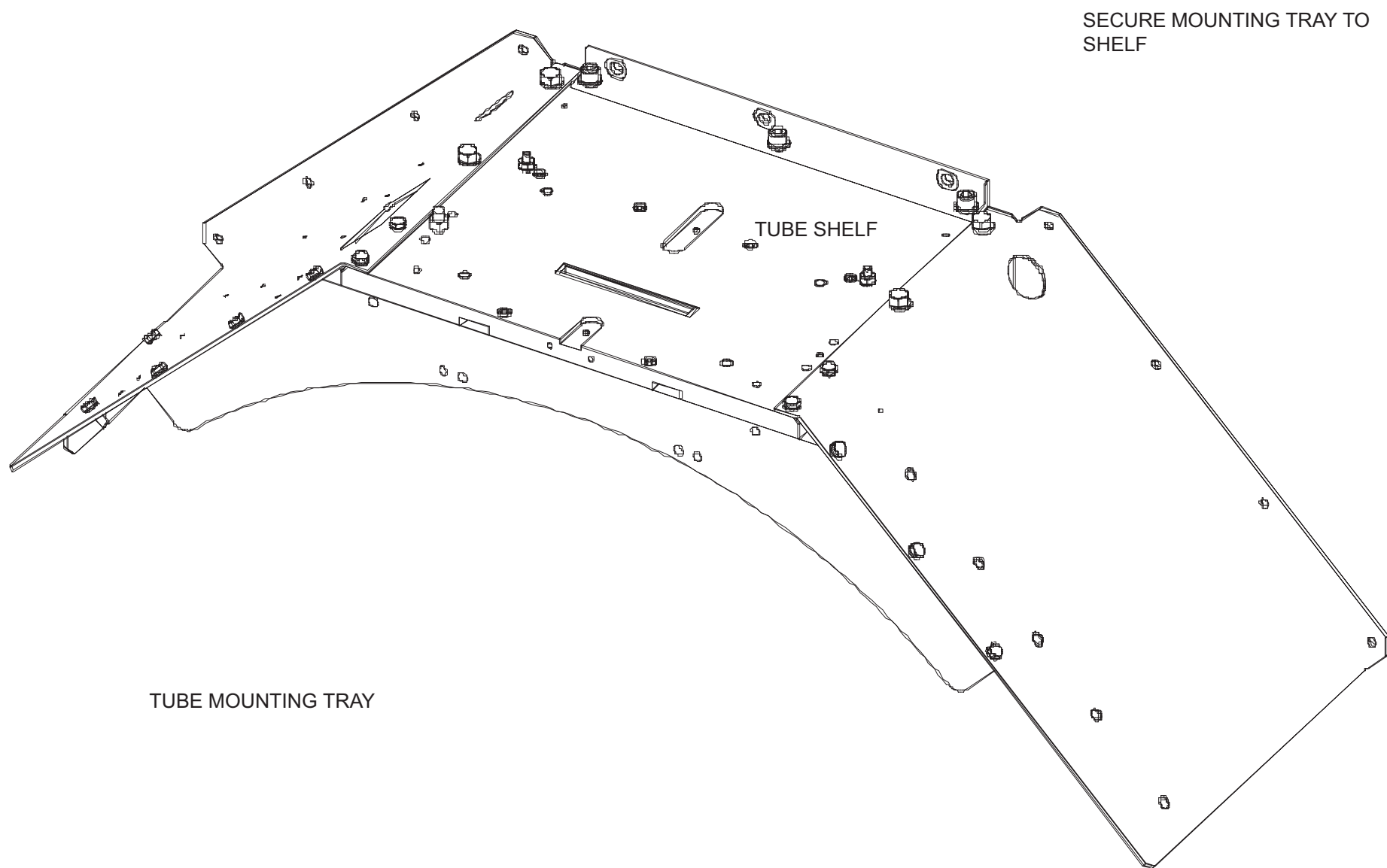


FIGURE 48 XRAY TUBE MOUNTING TRAY ON TUBE SHELF

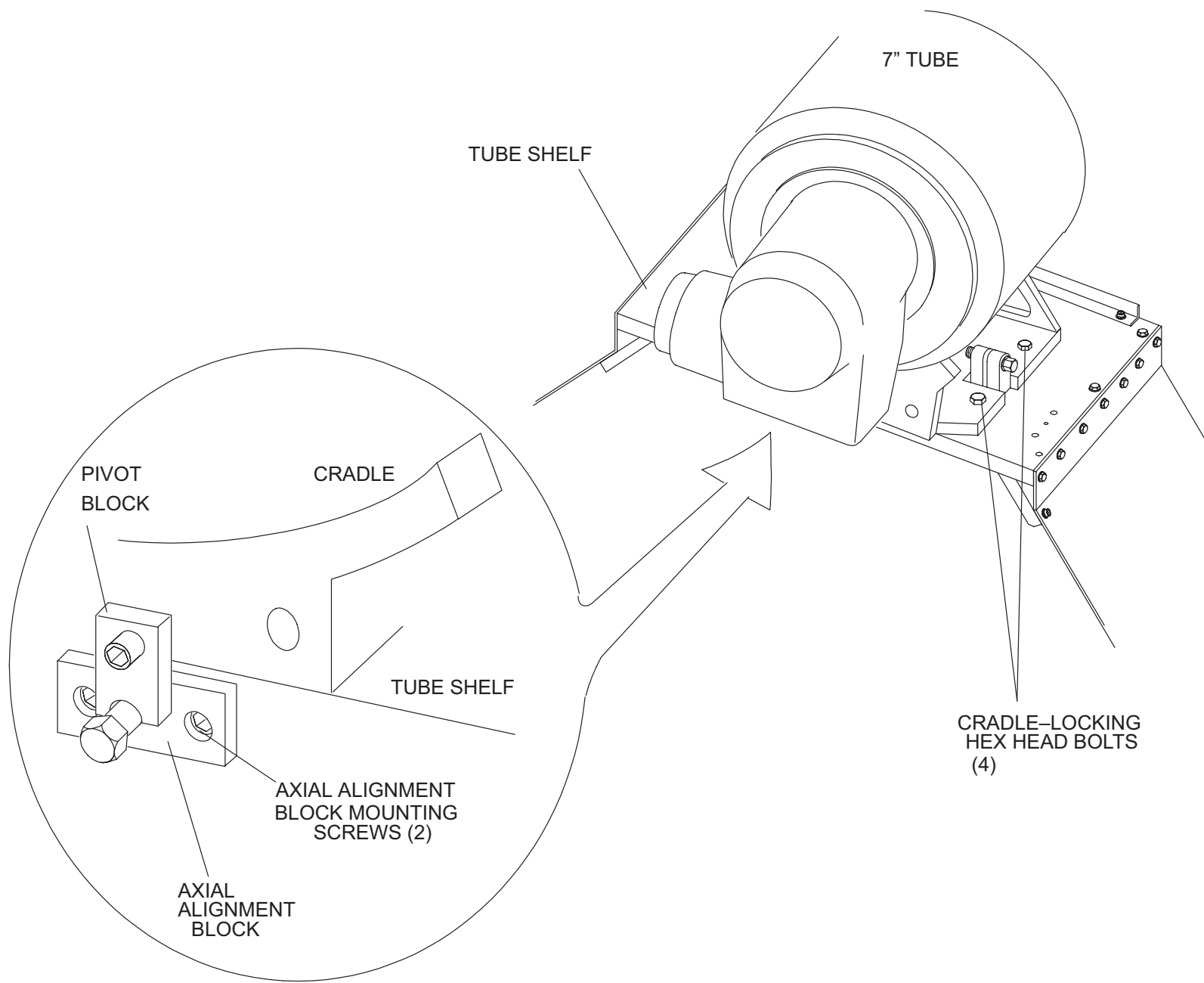


FIGURE 49 TUBE MOUNTED ON SHELF

NOTE

Recheck the mounting tray to ensure it is properly and securely installed.

5. Loosen one of the two Axial Alignment Block Mounting Socket Head Capscrews and remove the other. Let the Block swing down to allow removal of the X-Ray Tube Cradle. Refer to Figure 50.

NOTE

This method minimizes the need for axial adjustment of the X-Ray Tube if the *Axial Pivot block* is not loosened. However, later, care must be exercised not to damage the Patient Support with the protruding *Axial Pivot block*. Place a protective cover on the Carbon Top before setting the Tube and Cradle on it.

6. Install the XSC support rods into the scan frame so they can support the Heat Exchanger chassis.

Move Xray Tube Onto Patient Support

NOTE

The Couch Top must have a protective cover to prevent damage by the Axial Alignment Pivot Block attached to the tube cradle.

7. Place box with X-Ray Tube on left side of Patient Support. Open shipping container and remove packaging material.



WARNING

LIFTING HAZARD. THE HEAT EXCHANGER WEIGHS 45 LBS (20.5 KG) USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY CAN RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

8. Lift heat exchanger from container and place it on floor near front of Gantry.



WARNING

LIFTING HAZARD. THE X-RAY TUBE WEIGHS 83 LBS (37.7KG) IN THE CRADLE. USE EXTREME CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN SEVERE DAMAGE TO THE EQUIPMENT AND/OR INJURY TO PERSONNEL.

9. Place Lifting Straps through eyelets of Tube. Lift the X-Ray Tube from the shipping container and set it on the Patient Support with its emission port up.
10. With the X-Ray Tube Shutter facing upwards, remove protective covering. Check for the plastic covering over Shutter opening and check for dirt/dust. Remove any debris found on the plastic. **DO NOT REMOVE THE PLASTIC !**

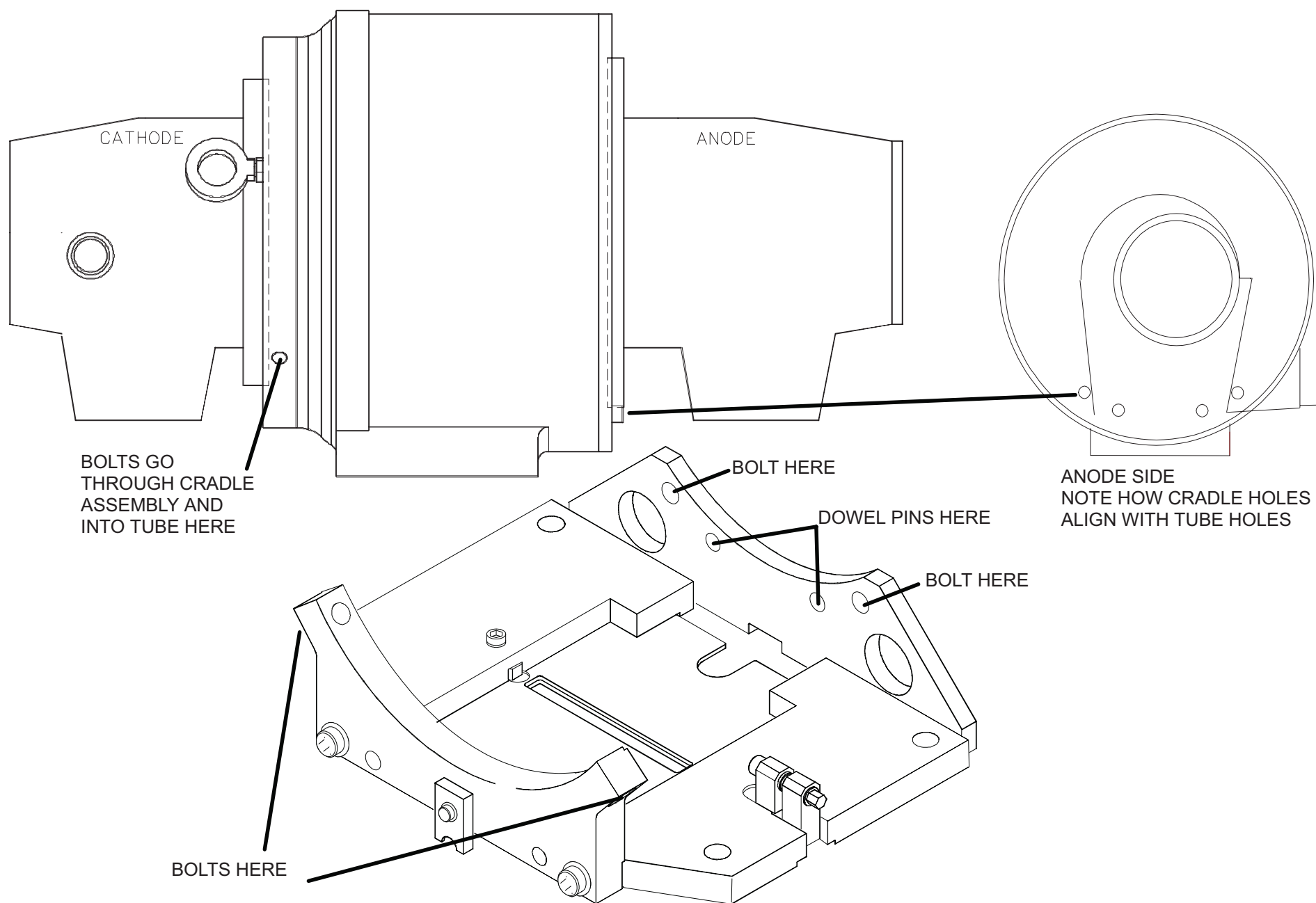


FIGURE 50 CRADLE ASSEMBLY

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11. Attach Cradle Assembly to Xray Tube. Follow illustration in Figure 50 for proper installation. Do not tighten Hex Head Bolts until Cradle End Plate is firmly seated on Tube and Base Plate Alignment Pins.
 - Torque Mounting Hex Head and Socket Head Screws to 15 ft–lbs.
12. Wipe off the X–Ray Tube with cloth and check for container–debris.
13. (If possible) Rotate the X–Ray tube so that the tube cradle side faces down and the oil hoses face left on the Patient support.
14. Remove the lifting straps from the eyelets and reposition as follows:
 - a. Reposition one strap through one eyelet, under the X–ray tube and up the other side through the other eyelet.
 - b. Route second strap through one cradle hole, under the x–ray tube and through hole on opposite side. Refer to Figure 51.

FIGURE NOTE: Figure 51 shows the tube on the cradle, but you route the straps in the same manner while the tube is on the Patient Support couch top.

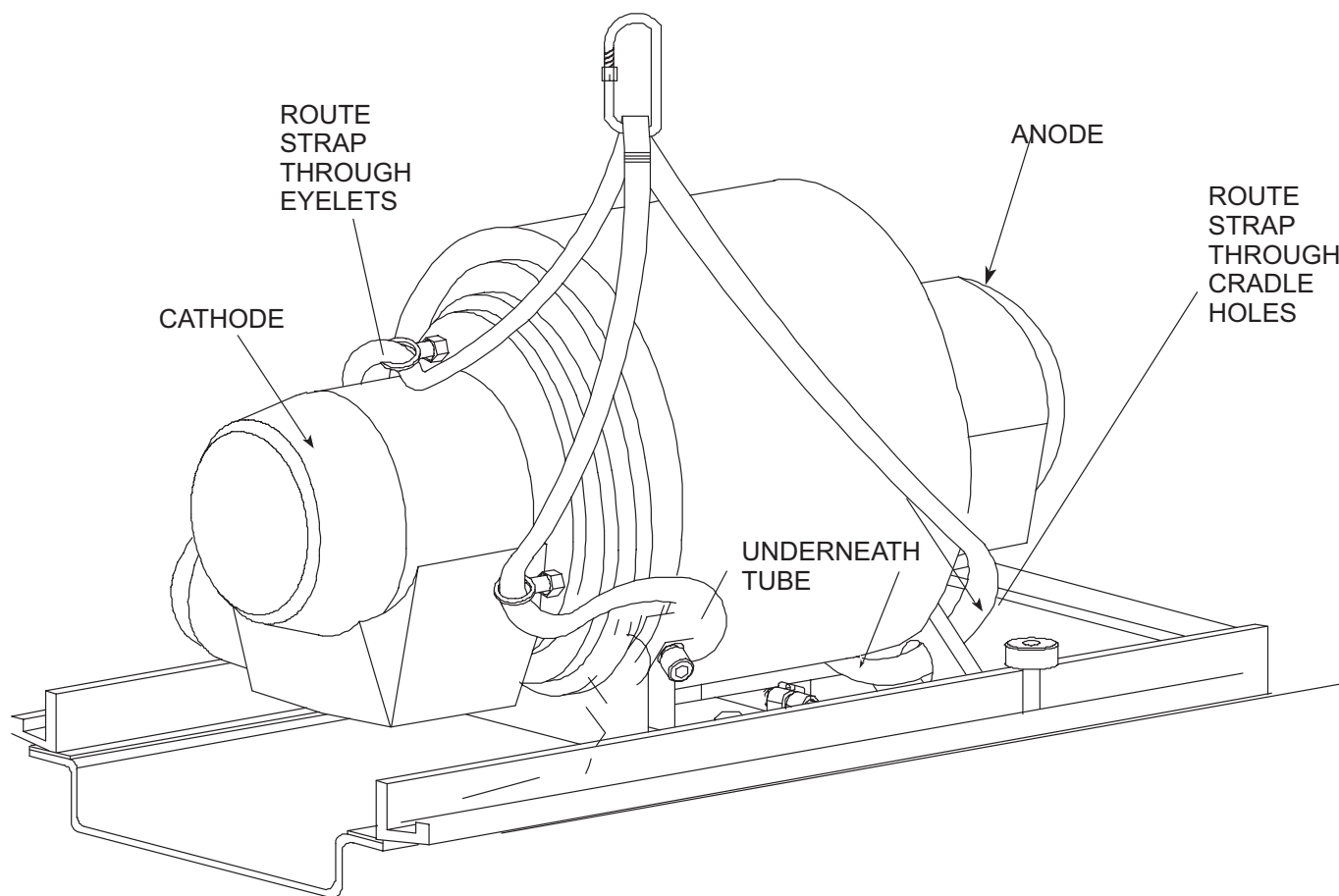


FIGURE 51 LIFTING STRAP ROUTING
MOUNTING TRAY

INSTALL TUBE INTO GANTRY

WARNING

CRUSH HAZARD. USE EXTREME CAUTION WHEN MOVING OR HANDLING OR IN GENERAL VICINITY OF X-RAY TUBE. SEE TUBE WEIGHTS IN THE TABLE AT FRONT OF THIS SECTION. FAILURE TO COMPLY CAN RESULT IN DAMAGE TO THE SCANNER OR X-RAY TUBE AND CAN RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.

15. Hoist the Xray Tube to the Guide Rails. Push Tube onto Rails until back of Cradle Assembly is flush with front of Tube Shelf.

WARNING

USE EXTREME CAUTION AND VERIFY THAT THE X-RAY TUBE IS FULLY SUPPORTED BY THE MOUNTING RAILS BEFORE REMOVING THE LIFTING STRAPS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

16. Remove Lifting Straps from around Xray Tube and move the Hoist Assembly out of the way.
17. Ensure that all cables are in the clear and that none will be pinched as you slide the tube into the cradle. Push the Xray Tube all the way back onto the Tube Shelf.
18. Replace the Cradle Locking Bolts and fasten the Cradle down enough to relieve pressure from the Mounting Rails.

19. Remove Mounting Rails and secure Axial Alignment Block in its original position with the Socket Head Capscrew removed earlier.
 - a. Apply Loc-Tite to Alignment Block Screws and tighten to hand-tight.
 - b. Tighten Cradle hardware to 15 ft-lbs.

**WARNING**

LIFTING AND CRUSH HAZARD. THE HEAT EXCHANGER WEIGHS 45 LBS. USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

INSTALL HEAT EXCHANGER INTO GANTRY

20. Lift the Heat Exchanger on XSC+ standoffs and push it back into position. Place the Cables behind the Heat Exchanger.

NOTE

Ensure the Cables rest on top the dogbone counterweights and that the cable-protecting rubber grommet is in place on top of the right heat exchanger mounting bracket.

21. Secure the Heat Exchanger to the scan frame. Refer to Figure 52.
22. Secure Oil Hoses with Hose Clamps. DO NOT OVERTIGHTEN HEX NUTS.
23. Connect Quick Disconnects J4 (Rotor Cable), J5 (Heat Exchanger Power Cable), and J6 (Thermal Switch Cable).

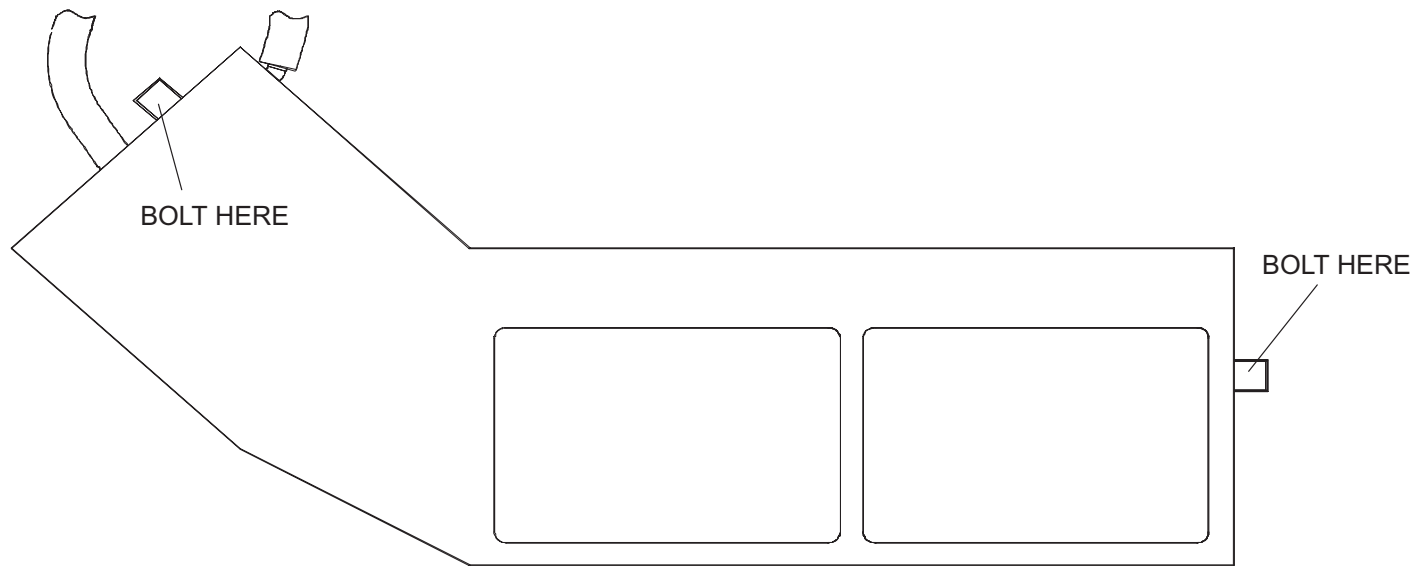


FIGURE 52 HEAT EXCHANGER MOUNTING

24. Install the Anode and Cathode High Voltage Federal Connectors. Refer to the Cable Cleaning and Installation procedure in the Repair manual.
25. VERIFY ALL CABLES AND HOSES ARE PROPERLY SECURED!
26. Remove the protective rubber strip over the detectors and secure it in the bottom of the Gantry. Remove the scan frame locking pin.
27. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.
28. Align the X-Ray Tube. Refer to the Xray Tube Mechanical Alignment procedure in the [MANEXP](#) section of the Diagnostic Gui manual.



CAUTION

ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY. CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES.

End of X-ray tube installation section. Continue to next page.